

Z-SENTINEL

Real-Time AI Fraud Risk & Decision Engine

Supporting Solution Design Document
IBM Z Datathon 2026 — Real-Time AI for Critical Decisions

Prototype positioning: This document describes a working prototype architecture and its intended alignment with IBM Z Transactional AI. Unless an IBM Z / Machine Learning for IBM z/OS environment is explicitly available to the team, IBM Z deployment is presented as the target integration architecture rather than a claim of completed production deployment.

1. Executive Summary

Z-Sentinel is a real-time AI-powered fraud risk and decision engine for financial transactions. The system is designed to analyze a transaction as it occurs, estimate its fraud probability, convert that prediction into a risk score, and recommend an immediate action: Approve, Review, or Block.

The prototype combines historical fraud data for model development with sandbox or synthetic transaction events for real-time demonstration. A machine learning layer produces the fraud probability and risk score; an explainability layer identifies important risk factors; and a dashboard presents the transaction, decision, explanation, and system performance.

The architecture is designed around the IBM Z Transactional AI concept: AI inference can be positioned close to mission-critical transaction processing, helping organizations make decisions at transaction time rather than relying only on post-transaction analysis. IBM identifies real-time fraud detection as a Transactional AI use case and describes Machine Learning for IBM z/OS as supporting in-transaction scoring through CICS/WOLA and REST interfaces.

2. Problem Statement

Financial institutions process very large numbers of transactions, while fraudulent activity can require a decision within the transaction workflow. Conventional rule-based controls and post-transaction analysis can miss emerging patterns or delay investigation.

The core problem is to analyze each incoming transaction quickly, identify abnormal or suspicious behavior, estimate fraud risk, and provide an actionable decision without unnecessarily separating the AI decision from the transaction-processing flow.

The solution must also be understandable: a fraud-risk prediction should provide useful reasons or contributing factors so that analysts can investigate high-risk transactions rather than receiving only an unexplained score.

3. Why Real-Time AI?

Fraud prevention is time-sensitive. Detecting a suspicious transaction after it has completed may reduce the opportunity to prevent loss. Z-Sentinel therefore treats each transaction as an event that can trigger immediate feature processing, model inference, risk evaluation, and an action recommendation.

IBM describes Transactional AI on IBM Z as embedding real-time predictive AI directly into transaction systems and specifically lists real-time fraud detection as a use case. Machine Learning for IBM z/OS provides real-time inference capabilities and interfaces intended for transactional applications.

4. Proposed Solution

Z-Sentinel receives a transaction event, validates and transforms the available transaction attributes, generates relevant behavioral features where data permits, and sends the feature vector to the fraud model.

The model returns a fraud probability. The risk engine converts that probability into a calibrated risk score and risk level. A configurable decision policy then recommends Approve, Review, or Block. For high-risk events, the system provides the leading model factors through explainability and can pass structured context to a GenAI layer for a human-readable investigation summary.

The prototype is deliberately modular so that the model, API, dashboard, and decision engine can be tested independently and later mapped to an IBM Z transactional environment.

5. System Architecture

Logical flow:

Transaction Event → Real-Time Feature Processing → ML Inference → Fraud Probability → Risk Scoring → Decision Engine → Action/Alert → Explainability → Dashboard

Prototype components: transaction simulator/sandbox event source, FastAPI service, preprocessing and feature-engineering layer, trained ML model, risk decision engine, SHAP-based explainability, database/log store, and Streamlit dashboard.

IBM Z target integration: the transaction application on IBM Z/z/OS can invoke a deployed ML scoring service through the supported transactional interfaces. The prototype's API contract is designed so that its inference layer can be adapted to that deployment model.

6. Data Flow

Historical data is used to train and evaluate the fraud model. For the real-time demonstration, incoming test/sandbox or synthetic transactions are transformed into the same feature schema expected by the model.

Where available, behavioral features can capture transaction velocity, deviation from a customer's normal amount, device novelty, location deviation, and unusual transaction timing. If a particular real-time source does not provide these attributes, the prototype will generate or simulate them rather than representing them as real customer data.

Data principle: the demonstration will use public, sandbox, or synthetic data and will not depend on real customer financial information or personally identifiable information.

7. ML Approach

The initial supervised-learning comparison will include Logistic Regression, Random Forest, and XGBoost. The final model will be selected using validation performance and practical considerations such as precision, recall, false-positive behavior, interpretability, and inference requirements.

The primary target is a binary fraud label where the selected training dataset provides a fraud indicator such as 0 = legitimate and 1 = fraud. Feature selection and preprocessing will be based on the finalized dataset rather than assuming that every behavioral feature is available in every source.

Because fraud datasets are typically imbalanced, the team will evaluate appropriate class-imbalance strategies and avoid relying on accuracy alone. Decision thresholds will be tuned using validation data and the relative cost of false positives and false negatives.

8. Real-Time Inference

The prototype exposes an inference contract that accepts a normalized transaction/feature payload and returns a structured prediction. The same contract can be used by the simulator, dashboard, or a future IBM Z integration adapter.

Example input: transaction_id, amount, merchant/category information, time, device/location indicators, and available behavioral features.

Example output: fraud_probability, risk_score, risk_level, decision, model_version, inference_time_ms, and top_risk_factors.

9. Risk Decision Engine

The risk engine separates machine-learning prediction from business action. This allows model thresholds and operational policies to be changed without retraining the model.

Illustrative flow: low-risk predictions → Approve; intermediate-risk predictions → Review; high-risk predictions → Block or enhanced verification. Exact thresholds will be determined from validation/calibration and the intended business cost trade-offs rather than being treated as universal values.

The system will log the prediction, decision, model version, timestamp, and relevant explanation fields for audit and evaluation.

10. Explainable AI

Z-Sentinel will use model explainability to identify the strongest contributing factors for a prediction. For tree-based models, SHAP can provide local feature contributions for individual transactions.

Example explanation: High transaction amount relative to recent customer behavior; unusually high transaction velocity; new device indicator.

GenAI, if included, will be used as an explanation and investigation-assistance layer rather than as the primary fraud classifier. The deterministic ML/risk pipeline remains responsible for the core prediction and decision.

11. IBM Z Integration

IBM states that Transactional AI on IBM Z can embed real-time predictive AI into transaction systems and identifies fraud detection as a relevant use case. IBM Machine Learning for IBM z/OS supports real-time inference through native CICS and WOLA interfaces and a RESTful interface.

In the target architecture, the transaction-processing application remains close to the transaction data and invokes AI scoring as part of the transaction flow. This reflects the IBM Z design principle of bringing AI inference close to mission-critical applications and data rather than requiring every transaction to be moved to a separate analytics environment.

Prototype boundary: if the team does not have access to an IBM Z / MLz environment during the datathon, the submission will demonstrate the equivalent transaction-time inference flow using the prototype stack and document the IBM Z/MLz integration point. It will not claim that the prototype is already deployed on IBM Z.

12. Dashboard

The dashboard will provide a real-time operational view of the system, including incoming transactions, fraud probability, risk score, risk level, decision, top risk factors, recent alerts, and model/system performance.

Suggested KPIs: total transactions processed, high-risk transactions, fraud predictions, approval/review/block counts, false-positive rate, precision/recall, average inference latency, and

throughput.

The dashboard is intended for demonstration and operational visibility; the authoritative transaction decision remains in the decision engine.

13. Security & Privacy

The prototype will avoid real customer financial data and personally identifiable information. Public datasets, sandbox/test transactions, and synthetic events will be used for development and demonstration.

In an IBM Z deployment, the design would rely on the security, governance, access-control, and operational capabilities of the enterprise environment. Sensitive transaction data should remain subject to the organization's security, privacy, retention, and compliance requirements.

Model and decision logs should be designed to support traceability while minimizing unnecessary sensitive data.

14. Evaluation Metrics

Model metrics: Precision, Recall, F1-score, ROC-AUC, confusion matrix, and—where useful—PR-AUC for imbalanced fraud data.

Decision metrics: false-positive rate, false-negative rate, review volume, and threshold sensitivity.

Real-time metrics: end-to-end inference latency, average/percentile latency, throughput, API response time, and successful event-processing rate.

Explainability: availability and consistency of top contributing risk factors for selected high-risk predictions.

15. Expected Impact

Z-Sentinel aims to demonstrate how AI can move fraud analysis from delayed reporting toward transaction-time decision support. The expected value is faster identification of suspicious events, prioritization of analyst attention, consistent risk scoring, and clearer explanations for high-risk predictions.

The prototype also demonstrates an architecture that can be adapted to IBM Z Transactional AI, where AI inference is positioned close to mission-critical transaction processing.

16. Future Scope

Future versions can extend the prototype with streaming platforms, richer customer behavioral profiles, online feature stores, model-drift monitoring, feedback from confirmed fraud outcomes, adaptive thresholding, stronger identity/device signals, and deeper IBM Z/MLz integration.

Additional models could support anomaly detection and ensemble approaches. A production implementation would also require formal security review, model governance, compliance validation, operational monitoring, and controlled deployment.

17. Prototype Technology Stack

Data: Python, Pandas, NumPy; public fraud dataset; sandbox/synthetic event data.

ML: Scikit-learn, Random Forest, Logistic Regression, XGBoost; SHAP for explainability.

Backend: FastAPI/REST API; database or event store for prototype transaction logs.

Dashboard: Streamlit with interactive visualizations.

Optional streaming: Kafka or another event-streaming layer if required by the final prototype; streaming infrastructure is not mandatory to demonstrate the core transaction-time inference concept.

IBM target: Machine Learning for IBM z/OS / Transactional AI integration architecture.

18. Dataset & Real-Time Data Strategy

Historical training: the team plans to use a public credit-card fraud dataset such as the Kaggle Credit Card Fraud Detection dataset for initial model development, subject to final team validation of licensing and feature suitability.

Real-time demonstration: sandbox/test payment events and/or synthetic transaction events will be used to simulate incoming transactions. This avoids dependence on real customer financial data.

Feature compatibility: the real-time event schema will be normalized to the model's training feature contract. Behavioral features will only be used when they are present in the source or explicitly simulated and documented.

Important: the final submission will state the exact dataset, source, target column, feature list, and real-time event source after the team freezes those implementation choices.

19. Conclusion

Z-Sentinel addresses the core requirement of Real-Time AI for Critical Decisions by combining transaction-level event processing, machine-learning fraud prediction, risk-based decisioning, explainability, and operational visualization.

The project is designed to demonstrate the central IBM Z Transactional AI idea: when a decision must be made at transaction time, AI inference should be positioned close to the transaction-processing workflow. The prototype therefore focuses not only on model accuracy, but also on the complete path from transaction arrival to immediate, explainable action.

Appendix A — Proposed ML/API Contract

Field	Purpose
transaction_id	Unique transaction/event identifier
fraud_probability	Model probability/score for fraud class
risk_score	Normalized operational risk score
risk_level	LOW / MEDIUM / HIGH
decision	APPROVE / REVIEW / BLOCK
top_risk_factors	Top explainability factors for the transaction
model_version	Version of the deployed model
inference_time_ms	Time taken for prediction
timestamp	Transaction/prediction timestamp

Appendix B — Key IBM References

- IBM AI on IBM Z — Transactional AI and real-time fraud detection:
<https://www.ibm.com/products/z/artificial-intelligence>
- IBM Machine Learning for IBM z/OS — real-time inference and interfaces:
<https://www.ibm.com/products/machine-learning-for-zos>
- IBM Redbooks — AI on IBM Z: Applications and Examples:
<https://www.redbooks.ibm.com/redpapers/pdfs/redp5758.pdf>
- IBM Redbooks — Turning Data into Insight with Machine Learning for IBM z/OS:
<https://www.redbooks.ibm.com/redbooks/pdfs/sg248552.pdf>